

The Triangle Inequality: A Proof Built from the Projection Picture

Lecture 1 notes

1 Statement

For any vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$,

$$\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|.$$

Equality holds exactly when one of $\mathbf{u}, \mathbf{v}$ is a nonnegative scalar multiple of the other — for nonzero vectors, when they point in the same direction. (Zero vectors cause no exception: if $\mathbf{u} = \mathbf{0}$ then $\mathbf{u} = 0 \cdot \mathbf{v}$, and the inequality is an equality trivially.)

2 Setup: decompose $\mathbf{v}$ using the projection picture

Assume $\mathbf{u} \neq \mathbf{0}$ so that “ $\mathbf{u}$ ’s direction” exists; if $\mathbf{u} = \mathbf{0}$ the inequality is immediate. Split $\mathbf{v}$ into two perpendicular pieces relative to $\mathbf{u}$:

- $\mathbf{v}_{\parallel}$ — the part of $\mathbf{v}$ parallel to $\mathbf{u}$ (the projection of $\mathbf{v}$ onto $\mathbf{u}$ ’s direction),
- $\mathbf{v}_{\perp}$ — the part perpendicular to $\mathbf{u}$ (the rejection).

These are orthogonal: $\mathbf{v}_{\parallel} \cdot \mathbf{v}_{\perp} = 0$.

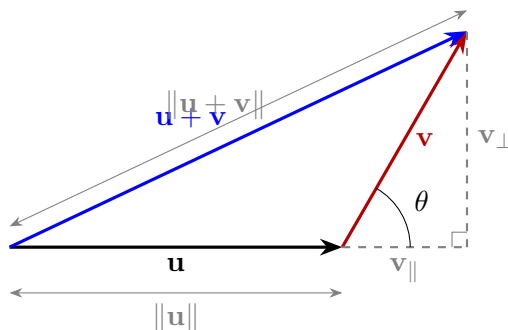


Figure 1: Head-to-tail drawing of $\mathbf{u} + \mathbf{v}$. Dashed lines show $\mathbf{v}$ split into its parallel part $\mathbf{v}_{\parallel}$ (extends $\mathbf{u}$ ’s line) and its perpendicular part $\mathbf{v}_{\perp}$ (lost sideways), so $\|\mathbf{u} + \mathbf{v}\|$ is the straight side of the triangle — shorter than walking $\mathbf{u}$ then $\mathbf{v}$ whenever $\theta > 0$.

3 Proof

Step 1 — split $\mathbf{u} + \mathbf{v}$ into perpendicular parts. Write

$$\mathbf{u} + \mathbf{v} = \underbrace{(\mathbf{u} + \mathbf{v}_{\parallel})}_{\parallel \text{ to } \mathbf{u}} + \underbrace{\mathbf{v}_{\perp}}_{\perp \text{ to } \mathbf{u}}.$$

The two brackets are perpendicular, so by Pythagoras,

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u} + \mathbf{v}_{\parallel}\|^2 + \|\mathbf{v}_{\perp}\|^2. \quad (1)$$

Step 2 — length of the first bracket. $\mathbf{u}$ and $\mathbf{v}_{\parallel}$ are collinear, so the length of their sum is at most the sum of their lengths (the triangle inequality in one dimension, which is trivial):

$$\|\mathbf{u} + \mathbf{v}_{\parallel}\| \leq \|\mathbf{u}\| + \|\mathbf{v}_{\parallel}\|. \quad (2)$$

(If $\mathbf{u} \cdot \mathbf{v} \geq 0$, the projection is a nonnegative multiple of $\mathbf{u}$ and we have equality; if the angle is obtuse, it is a negative multiple and the left side is even smaller. Either way the inequality holds.)

Step 3 — a fact about $\mathbf{v}$ itself. Because $\mathbf{v}_{\parallel} \perp \mathbf{v}_{\perp}$, Pythagoras on $\mathbf{v}$ gives

$$\|\mathbf{v}\|^2 = \|\mathbf{v}_{\parallel}\|^2 + \|\mathbf{v}_{\perp}\|^2. \quad (3)$$

Step 4 — the key inequality. A projection is never longer than the vector it comes from:

$$\|\mathbf{v}_{\parallel}\| \leq \|\mathbf{v}\|. \quad (4)$$

(Geometrically: $\|\mathbf{v}_{\parallel}\| = \|\mathbf{v}\| |\cos \theta| \leq \|\mathbf{v}\|$.)

Now combine (1) with (2), expand, and substitute (3):

$$\begin{aligned} \|\mathbf{u} + \mathbf{v}\|^2 &\leq (\|\mathbf{u}\| + \|\mathbf{v}_{\parallel}\|)^2 + \|\mathbf{v}_{\perp}\|^2 \\ &= \|\mathbf{u}\|^2 + 2\|\mathbf{u}\|\|\mathbf{v}_{\parallel}\| + \|\mathbf{v}_{\parallel}\|^2 + \|\mathbf{v}_{\perp}\|^2 \\ &= \|\mathbf{u}\|^2 + 2\|\mathbf{u}\|\|\mathbf{v}_{\parallel}\| + \|\mathbf{v}\|^2 \\ &\leq \|\mathbf{u}\|^2 + 2\|\mathbf{u}\|\|\mathbf{v}\| + \|\mathbf{v}\|^2 \quad \text{by (4)} \\ &= (\|\mathbf{u}\| + \|\mathbf{v}\|)^2. \end{aligned}$$

Step 5 — take square roots. Both sides are non-negative, so

$$\boxed{\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|}.$$

4 When is it equality?

The chain contains *two* inequalities, and both must collapse to equalities:

- (2) is an equality iff the projection $\mathbf{v}_{\parallel}$ is a nonnegative multiple of $\mathbf{u}$, i.e. $\mathbf{u} \cdot \mathbf{v} \geq 0$ (same direction).
- (4) is an equality iff $\mathbf{v}_{\perp} = \mathbf{0}$ (no perpendicular part, i.e. $\mathbf{v}$ parallel to $\mathbf{u}$).

Both hold at once iff $\mathbf{v} = c\mathbf{u}$ for some $c \geq 0$: parallel alone is not enough, since $\mathbf{v} = -\mathbf{u}$ has no perpendicular part yet gives $\|\mathbf{u} + \mathbf{v}\| = 0 \neq 2\|\mathbf{u}\| = \|\mathbf{u}\| + \|\mathbf{v}\|$. This matches the geometric picture: the head-to-tail drawing is a straight line exactly when both vectors share a direction.

5 The engine underneath: the dot product and Cauchy–Schwarz

The key inequality (4) actually needs no new machinery: squaring (3) and dropping the non-negative term $\|\mathbf{v}_\perp\|^2 \geq 0$ gives it in one line. What does need the dot product is computing $\mathbf{v}_\parallel$ itself. Define it algebraically as the projection formula

$$\mathbf{v}_\parallel = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\|^2} \mathbf{u}.$$

Then its length is a computation:

$$\|\mathbf{v}_\parallel\| = \frac{|\mathbf{u} \cdot \mathbf{v}|}{\|\mathbf{u}\|},$$

and combining with (4) yields

$$|\mathbf{u} \cdot \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|,$$

the **Cauchy–Schwarz inequality**, now a genuine corollary of the setup rather than an assumption in disguise. Geometrically it says $|\cos \theta| = |\mathbf{u} \cdot \mathbf{v}| / (\|\mathbf{u}\| \|\mathbf{v}\|) \leq 1$ — which is exactly why the angle between vectors is well-defined, and why the course’s picture $\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$ is consistent.

So the triangle inequality is, at bottom, a consequence of the dot product — which is why Lecture 2 promotes the dot product (the inner product) to the central tool, and why the norm is defined the way it is. Everything connects.